MATH 127 Calculus for the Sciences

Lecture 34

November 25, 2025

Today's lecture

Last time
Integration by substitution

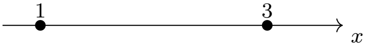
This time
Course note coverage Sections 5.2.1 and 5.2.2
Sigma notation
Riemann sums again, but using sigma notation

Information
Quiz 10 today!

Motivation

- What does the definite integral $\int_1^3 x^2\,dx$ represent?
- How do we do a left Riemann approximation this integral by splitting $[1,3]$ into 4 pieces?

1. Mark the splitting of the interval on the x-axis.



2. Use the formula

$$\text{Area} = \text{sum of } \left(\boxed{\qquad\qquad} \times \boxed{\qquad\qquad}\right) \text{ for each piece}$$
$$= \text{in this case we are adding } \boxed{\quad} \text{ terms}$$

Question Writing like this is ok, but what if you need to split into 100 pieces? There's not enough space to add 100 things!

Sigma notation

Notation. We write

$$\sum_{i=m}^{n} f(i)$$

to mean we *add up* the values $f(i)$ for all integers i starting at m and going up to and including n.

Example. So if I want to add together $f(3)$, $f(4)$, all the way to $f(30)$, then I can write

$$\sum_{i=3}^{30} f(i) = f(3) + f(4) + f(5) + \cdots + f(30).$$

Instead of the right hand side which is super long, this notation is easier to write and easier to read.

Example 1

Example. Expand the following

$$\sum_{i=2}^{5} \left(i^2 + e^i\right)$$

$$\begin{aligned}\sum_{K=2}^{5} K! &= \boxed{} + \boxed{} + \boxed{} + \boxed{} \\ &= \boxed{} + \boxed{} + \boxed{} + \boxed{} \\ &= \boxed{}\end{aligned}$$

Example 1 (cont.)

Example. Expand the following

$$\sum_{i=0}^{n-1} f(x_i)\Delta x$$

$$\sum_{i=0}^{n-1} f(x_i)\Delta x = \boxed{} + \boxed{} + \cdots + \boxed{}.$$

Example 2

Exercise. Write this in sigma notation

$$0.5 + 1 + 1.5 + 2 + 2.5$$

Hint. There are 5 things to add, so maybe it should be

$$\sum_{i=1}^{5} \boxed{}$$

Summation properties

Summation properties. For any real number c:

1. $\displaystyle\sum_{i=m}^{n} c\, f(i) = c \sum_{i=m}^{n} f(i)$
2. $\displaystyle\sum_{i=m}^{n} \big(f(i) + g(i)\big) = \sum_{i=m}^{n} f(i) + \sum_{i=m}^{n} g(i)$

In words, you can say

- multiplying a number to a sum distributes that number to each summand;
- adding two sums together is the same as summing all of them all at once.

Summation formulas

Summation formulas. For positive integer n:

$$\sum_{i=1}^{n} 1 = n,$$

$$\sum_{i=1}^{n} i = \frac{n(n+1)}{2},$$

$$\sum_{i=1}^{n} i^2 = \frac{n(n+1)(2n+1)}{6}.$$

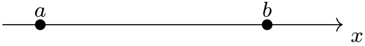
Gauss proved the formula above in elementary school when he was forced to add $1 + 2 + 3 + \cdots + 100$.

Today you can use his findings and conclude

$$1 + 2 + \cdots + 100 = \sum_{i=1}^{100} \boxed{} = \boxed{} = \boxed{}$$

Right Riemann sums

Let's split $[a, b]$ into n pieces.



The length of each piece we denote Δx, to represent change in x. How long is this length?

$$\Delta x = \boxed{}$$

Let's say the first piece is $[x_0, x_1]$, the second piece is $[x_1, x_2]$, the third piece is $[x_2, x_3]$, all the way to the n-th piece, which would be denoted as $\boxed{}$.
This means $x_0 = a$, $x_1 = a + \Delta x$, and

$$x_2 = a + \boxed{}, \ldots, x_n = a + \boxed{}$$

Right Riemann sums

Now the **right** Riemann sum says

$$\begin{aligned}
\text{Area} &= \text{sum of } (\text{base length} \times \text{output of } f(x) \text{ on the right side}) \text{ for each piece} \\
&= \text{sum of } (\Delta x \times f(x_i)) \text{ for the } i\text{-th piece} \\
&= \text{sum of } (\Delta x \times f(a + i\Delta x)) \text{ for the } i\text{-th piece} \\
&= \sum_{i=1}^{n} f(a + i\Delta x) \cdot \Delta x
\end{aligned}$$

The right hand side is a formula without using english words. So even aliens from Mars can understand!

Remark What would the sum be if you do **left** Riemann sum?
Based on the above formula, which do you like more, left or right Riemann?

Example 3

Example. Use the right Riemann sum to approximate the area under

$$f(x) = 2x \quad \text{for } 0 \leq x \leq 1$$

using $n = 100$ subdivisions.

We have

$\Delta x = \text{length of each piece} = \boxed{}$ $\quad x_i = \text{right side value of each piece} = \boxed{}$

Then

$$\begin{aligned} \text{Area} &\approx \sum_{i=1}^{100} f(x_i)\,\Delta x \\ &= \sum_{i=1}^{100} 2\left(\boxed{}\right) \cdot \boxed{} \\ &= \sum_{i=1}^{100} \boxed{} \\ &= \boxed{} \cdot \boxed{} \leftarrow \text{thanks to Gauss} \\ &= \boxed{} \end{aligned}$$

The real survey

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